Natalyx MVP Fixes

Work Package 12: Ask/Learn shell and mobile cutover

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1 Work Package 12: Ask/Learn shell and mobile cutover

Review draft only. Ownership and sequencing remain recommendations until Luke and Alan approve the plan together.

Plan	Value
Work package	WP12
Plan file	`WP12 - Ask and Learn shell and mobile cutover.md`
Proposed PRs	P12-01, P12-02, P12-03
Inspection snapshot	2026-07-10 20:12 America/New_York

1.1 Purpose and outcome

Problem. Participant Ask/Learn entry points, case shell, legacy routes, and mobile interactions are not yet one coherent clickable journey with explicit context boundaries.

Why this package exists. It creates a reviewable boundary so the programme can progress without mixing product, data, security, infrastructure and rollout concerns in one oversized change.

Completed outcome. A unified scope-labelled Ask/Learn shell, accessible mobile participant journey, and verified redirect-based legacy cutover with no broken or misleading controls.

Explicit non-goals.

- No final content authoring
- No staff/operator redesign

- No broad visual rebrand
- No removal before telemetry proves safe

1.2 Verified current-state evidence

Classification is explicit: **verified** is observed code/PR/live behavior; **verified limitation** records an evidence gap; **design authority** is an approved planning source; **design decision** is proposed future behavior. None is silently treated as another.

ID	Class	Repository / ref	File, construct or location	Finding
E-006	verified	app / PR #122@e853821	PR comment Review round 2 addressed and changed workflow/tests	Updated PR #122 reports 499 tests plus return_to, TTL, CAS, self-invite, lens and Learn fail-closed fixes; green checks were observed. These are salvage candidates, not blanket merge approval.
E-012	verified	app / PR #122@e853821	app/storage/invitations.py and PR review-round comment	The updated case invitation work adds 14-day ceiling, CAS transitions, mismatch confirmation, self-invite refusal, return_to preservation and paired lens tests; it has no Node 0 workspace target.
E-024	verified	design-system / main@43dab986; PR #7@cedfcc4; PR #8@4656426	components, tokens, Jinja attrs/parity guidance	Participant DS recipes and parity guidance exist; local brand assets are also dirty. Internal/operator DS is a separate repository and boundary.

1.3 Architecture and contracts

- The shell labels general, private, and shared-case context and never implies a source exists when it does not.
- Server owns state and permissions; client code enhances interaction only.
- Legacy routes redirect through allowlisted, telemetry-observed compatibility paths before deletion.

Contract area	Package rule
Ownership boundary	`app`; repository-specific PRs remain separate.
Entities/state	No new runtime entity; evidence/coordination only.
Authorization	Authenticated subject/service plus entity membership, role, purpose, consent and expected revision; query parameters, links and UI state are never authority.
API/service	Pass versioned scoped envelopes only. Stable redirects with query sanitization.
Storage	No new storage; preserve current state read-only for this package.

Contract area	Package rule
Audit	Use the common transition envelope and append-only source/event identifiers; redact tokens, raw documents, prompts and sensitive payloads.
Failure	Fail closed for authorization, provenance, money, document and PHI boundaries; expose a bounded actionable error and preserve retry/repair evidence.
Observability	Correlation, policy/schema/config version, transition and dependency result metrics without PHI/PII payload logging.
Compatibility	Additive/dual-compatible changes, explicit version pins, old/new revision window and named forward repair.
Privacy/PHI	Synthetic only through A2; approved PII only at B1; PHI remains blocked until formal B2.

1.4 Explicit PR sequence

Order	ID	Title	Repository	Dependencies	Gate	Size	Readiness
1	P12-01	Unify Ask and Learn shell with explicit context scopes	app	P10-03, P11-04, P09-04	A2	Large	Dependencies unmet
2	P12-02	Complete mobile participant interaction and accessibility cutover	app	P05-03, P06-03, P09-04, P12-01	A2	Exceptional	Dependencies unmet
3	P12-03	Retire legacy participant routes behind verified redirects	app	P12-02, P13-04	B1	Large	Dependencies unmet

1.4.1 P12-01 - Unify Ask and Learn shell with explicit context scopes

Field	Plan
Repository / base	`app` (`main`)
Gate / size / readiness	A2 / Large / Dependencies unmet
Dependencies	P10-03, P11-04, P09-04
Blocking decisions/checkpoints	D-AI-01
Source / salvage	Current Ask widget/Learn workspace, PR #124

1.4.1.1 Objective Present general Learn, private personalization, and approved shared-case Ask as visibly distinct contexts with source cards and safe switching.

1.4.1.2 Scope

- Context scope badge/selector
- Suggested questions and citations
- Boundary and unavailable states
- Case-aware authorization

1.4.1.3 Explicit non-goals

- No context merge
- No case fact write

1.4.1.4 Expected files or modules

- `app/ask_natalyx.py`
- `app/templates/ask_natalyx*.html`
- `app/templates/learn_workspace.html`
- `app/static/js/learn_chat.js`
- `tests/python/test_ask_learn_shell.py`

1.4.1.5 Change map

Area	Planned change
Schema/model	None in this PR; any later schema change stays in its named migration PR.
API/routes	Pass versioned scoped envelopes only.
UI/product surface	Unified but scope-explicit shell.
Infrastructure/workflow	No cloud/infrastructure resource or workflow change.
Migration/backfill	No migration or backfill.
Security/privacy	Preserve existing fail-closed boundaries; validate the PR-specific privacy/security cases.
Architecture/docs/config	Document the contract, version pins and safe partial-merge behavior.

1.4.1.6 Tests and validation

Objective tests:

- General/private/shared views
- wrong-case denial
- logout/login
- empty/failure
- mobile/keyboard/screen-reader

Local validation:

- `python -m unittest discover -s tests/python`
- Run the focused test module(s) named above against SQLite and PostgreSQL where applicable.

Cloud validation: run only through Gate **A2** on a known revision, using synthetic data until B1 and no PHI until B2. A merge or green deployment is not gate acceptance.

1.4.1.7 Diagnostics and observability Emit correlation/idempotency/policy/version identifiers, transition/result counters and bounded failure reasons; never log tokens, processor payloads, raw documents, prompts, PHI or unnecessary PII.

1.4.1.8 Rollout Deploy behind the narrowest practical feature/config gate, verify the named synthetic flow and diagnostics on a known revision, then widen only at the associated programme gate.

1.4.1.9 Rollback or forward repair Disable the feature or return to the prior compatible image; retain append-only audit/ledger history and repair with a new event/revision rather than destructive mutation.

1.4.1.10 Dependencies and parallel work Hard dependencies: P10-03, P11-04, P09-04. Parallel opportunities: P12-03

1.4.1.11 Objective acceptance criteria

- ☐ Present general Learn, private personalization, and approved shared-case Ask as visibly distinct contexts with source cards and safe switching.
- ☐ Every listed objective test passes locally and at the required gate where applicable.
- ☐ Every explicit non-goal remains out of the diff and no protected worktree/branch is modified during extraction.
- ☐ Rollout/repair evidence is attached before merge; unresolved decisions remain blocked rather than encoded as guesses.

1.4.1.12 Ownership and review

Role	Recommendation
DRI	Luke
Implementer	Luke (proposed; pending joint approval)
Reviewer	Alan
Final approver	Joint
Required expertise	privacy UX, FastAPI/Jinja, accessibility
Higher-capability review	Mandatory before merge

1.4.2 P12-02 - Complete mobile participant interaction and accessibility cutover

Field	Plan
Repository / base	`app` (`main`)
Gate / size / readiness	A2 / Exceptional / Dependencies unmet
Dependencies	P05-03, P06-03, P09-04, P12-01
Blocking decisions/checkpoints	None beyond dependencies and plan approval
Source / salvage	PR #122 mobile journey, PR #124 Node 0, DS PR #8

1.4.2.1 Objective

Run a bounded participant-only mobile/accessibility pass across entry, auth, Node 0, GC claim, case, documents, Ask, and Learn using the released DS contract.

1.4.2.2 Scope

- Navigation/forms/cards/dialogs
- 320/375/768 widths
- Keyboard/focus/reduced motion
- Error/loading/empty states

1.4.2.3 Explicit non-goals

- No internal/operator surface
- No visual rebrand
- No zero-hex purity gate

1.4.2.4 Expected files or modules

- `app/templates/participant*/`
- `app/templates/node0/`
- `app/templates/case/`
- `app/static/participant*/`
- `tests/browser/participant_mobile.spec.*`

1.4.2.5 Change map

Area	Planned change
Schema/model	None in this PR; any later schema change stays in its named migration PR.
API/routes	No public API/route contract change.
UI/product surface	Participant interaction defects and concrete DS adoption only.
Infrastructure/workflow	No cloud/infrastructure resource or workflow change.
Migration/backfill	No migration or backfill.
Security/privacy	Preserve existing fail-closed boundaries; validate the PR-specific privacy/security cases.
Architecture/docs/config	Document the contract, version pins and safe partial-merge behavior.

1.4.2.6 Tests and validation

Objective tests:

- All required clickable journeys
- axe/manual keyboard
- reduced motion
- orientation/zoom
- slow dependency

Local validation:

- `python -m unittest discover -s tests/python`
- Run the focused test module(s) named above against SQLite and PostgreSQL where applicable.

Cloud validation: run only through Gate **A2** on a known revision, using synthetic data until B1 and no PHI until B2. A merge or green deployment is not gate acceptance.

1.4.2.7 Diagnostics and observability Emit correlation/idempotency/policy/version identifiers, transition/result counters and bounded failure reasons; never log tokens, processor payloads, raw documents, prompts, PHI or unnecessary PII.

1.4.2.8 Rollout Deploy behind the narrowest practical feature/config gate, verify the named synthetic flow and diagnostics on a known revision, then widen only at the associated programme gate.

1.4.2.9 Rollback or forward repair Disable the feature or return to the prior compatible image; retain append-only audit/ledger history and repair with a new event/revision rather than destructive mutation.

1.4.2.10 Dependencies and parallel work Hard dependencies: P05-03, P06-03, P09-04, P12-01. Parallel opportunities: None inside this package beyond review/test preparation.

1.4.2.11 Objective acceptance criteria

- ☐ Run a bounded participant-only mobile/accessibility pass across entry, auth, Node 0, GC claim, case, documents, Ask, and Learn using the released DS contract.
- ☐ Every listed objective test passes locally and at the required gate where applicable.
- ☐ Every explicit non-goal remains out of the diff and no protected worktree/branch is modified during extraction.
- ☐ Rollout/repair evidence is attached before merge; unresolved decisions remain blocked rather than encoded as guesses.

1.4.2.12 Ownership and review

Role	Recommendation
DRI	Luke
Implementer	Luke (proposed; pending joint approval)
Reviewer	Alan
Final approver	Joint
Required expertise	responsive UI, accessibility, browser testing
Higher-capability review	Risk-proportionate; not mandatory by programme default

1.4.3 P12-03 - Retire legacy participant routes behind verified redirects

Field	Plan
Repository / base	`app` (`main`)
Gate / size / readiness	B1 / Large / Dependencies unmet
Dependencies	P12-02, P13-04
Blocking decisions/checkpoints	None beyond dependencies and plan approval
Source / salvage	Current route inventory, PRs #112/#119 planning maps

1.4.3.1 Objective Move legacy participant entry/case/Ask/Learn routes behind allowlisted redirects and telemetry, then remove only after a measured compatibility window.

1.4.3.2 Scope

- Route inventory and redirect map
- Bookmark/return_to compatibility
- Telemetry and rollback flag
- Later dead-code removal

1.4.3.3 Explicit non-goals

- No immediate destructive deletion
- No operator route change

1.4.3.4 Expected files or modules

- `app/routes/participant*.py`
- `app/main.py`
- `tests/python/route_inventory.json`
- `docs/runbooks/legacy-cutover.md`

1.4.3.5 Change map

Area	Planned change
Schema/model	None in this PR; any later schema change stays in its named migration PR.
API/routes	Stable redirects with query sanitization.
UI/product surface	No visible participant/operator UI change.
Infrastructure/workflow	No cloud/infrastructure resource or workflow change.
Migration/backfill	No migration or backfill.
Security/privacy	Preserve existing fail-closed boundaries; validate the PR-specific privacy/security cases.
Architecture/docs/config	Document the contract, version pins and safe partial-merge behavior.

1.4.3.6 Tests and validation

Objective tests:

- Every legacy route
- query/open-redirect safety
- stale bookmarks
- rollback flag
- route parity

Local validation:

- `python -m unittest discover -s tests/python`
- Run the focused test module(s) named above against SQLite and PostgreSQL where applicable.

Cloud validation: run only through Gate **B1** on a known revision, using synthetic data until B1 and no PHI until B2. A merge or green deployment is not gate acceptance.

1.4.3.7 Diagnostics and observability

Legacy hit counters and unknown-route alerts.

1.4.3.8 Rollout

Deploy behind the narrowest practical feature/config gate, verify the named synthetic flow and diagnostics on a known revision, then widen only at the associated programme gate.

1.4.3.9 Rollback or forward repair

Disable the feature or return to the prior compatible image; retain append-only audit/ledger history and repair with a new event/revision rather than destructive mutation.

1.4.3.10 Dependencies and parallel work

Hard dependencies: P12-02, P13-04. Parallel opportunities: P12-01

1.4.3.11 Objective acceptance criteria

- ☐ Move legacy participant entry/case/Ask/Learn routes behind allowlisted redirects and telemetry, then remove only after a measured compatibility window.
- ☐ Every listed objective test passes locally and at the required gate where applicable.
- ☐ Every explicit non-goal remains out of the diff and no protected worktree/branch is modified during extraction.
- ☐ Rollout/repair evidence is attached before merge; unresolved decisions remain blocked rather than encoded as guesses.

1.4.3.12 Ownership and review

Role	Recommendation
DRI	Luke
Implementer	Luke (proposed; pending joint approval)

Role	Recommendation
Reviewer	Alan
Final approver	Joint
Required expertise	Python/FastAPI, PostgreSQL, security-aware product engineering
Higher-capability review	Risk-proportionate; not mandatory by programme default

1.5 Cross-repository coordination

Repository	Authority after merge
`app`	routing, validation, authorization, participant/case/workflow truth, PHI policy and commercial domain

Merge order inside this package: `P12-01` → `P12-02` → `P12-03`.

Handoff rule: producer contract/version must merge and be pinned before its consumer enables the feature. During a partial merge, consumers fail closed or retain the last approved version; they do not fabricate data or silently broaden access.

Compatibility window: keep prior readers/writers and rollback image usable through the named gate; contract/migration PRs document the exact removal point.

1.6 Test and acceptance matrix

Test dimension	Method	Objective result
Happy path	Clickable IP/GC flows; Context labels; No-JS/reduced-motion/keyboard; Mobile widths; Redirect and stale bookmark; Logout/login/resume	Expected state/event/UI and named evidence are produced once.
Authorization failure	Wrong/no subject, role, membership, purpose or consent	403/404 as appropriate; no write or data leak; audit safe failure.
Role differences	Separate IP, GC, operator, provider/service identities	Only the approved lens/projection/actions appear.
Retries/duplicates	Repeat same canonical request/event	Original result returned; no duplicate membership/case/invoice/document/event.
Concurrency	Two writers/claimants/migrators/workers	One valid transition; loser receives deterministic stale/conflict response.
Stale state	Old revision/config/terms/token/artifact	Fail closed or approved compatibility behavior; never silent coercion.
Malformed input	Boundary, encoding, size, enum, redirect and identifier corpus	Validation before side effects; safe error.
Unavailable dependency	Database, intelligence, processor, S3/scanner or auth unavailable	Explicit degraded/blocked behavior; no hallucinated success.

Test dimension	Method	Objective result
Partial failure	Crash between external effect, DB transaction and outbox	Retry/forward repair converges from durable evidence.
Logout/login/resume	Session expires/restarts mid-flow	Return to authorized current state; no query/link authority.
Mobile/accessibility	320/375/768 widths, keyboard, focus, reduced motion	Core task completes with truthful state and no clipped controls.
Synthetic cloud	Known dev revision and synthetic identities/data	Gate artifact records SHA, dependency status, audit and cleanup.
Audit/observability	Inspect events/metrics/logs	Versions, actor/authority, correlation and reason present; no sensitive payload.

1.7 Risks and decisions

Type	ID / risk	Owner	Affected PRs / latest point	Treatment
Blocking/product/compliance decision	D-AI-01 - Approve scopes, allowed sources, model policy, proposal reviewers, class-5 evidence rail and degraded-mode copy.	Luke	P10-01, P10-02, P10-03, P12-01; Before P10-01	D-AI-01 - Generic case members cannot create reviewed/provider truth; intelligence remains advisory and scope explicit.
Implementation risk	Context ambiguity	Luke or named PR DRI	Package PRs; before associated gate	Mitigate through the explicit tests, rollout and forward-repair plan
Implementation risk	Mobile-only dead ends	Luke or named PR DRI	Package PRs; before associated gate	Mitigate through the explicit tests, rollout and forward-repair plan
Implementation risk	Premature legacy removal	Luke or named PR DRI	Package PRs; before associated gate	Mitigate through the explicit tests, rollout and forward-repair plan
Implementation risk	Design drift	Luke or named PR DRI	Package PRs; before associated gate	Mitigate through the explicit tests, rollout and forward-repair plan

1.8 Existing-work salvage

Source	Head	Valuable content	Destination	Recommendation	Extraction status/safety
app#112	9605d88e31bd	Source-first 81-node/12-cluster surface map, participant-renderer boundaries, wave plan and cross-repo seams.	P12-03	Selective port, then supersede after extraction	Catalogued; content remains to be reconciled into replacement PR acceptance tests.
app#119	a8cd2f428535	Concise product-leverage framing and boundaries for the MVP flow.	P12-03	Retain as source or merge separately after plan decision	Catalogued; reconcile terminology into approved programme.
app#121	529d377e2337	Primary Node 0 participant model/UI, PDF/guide presentation and focused surface tests.	P12-02	Selective UI/test port, then supersede	Unextracted; #124 may contain later copies but equivalence must be proved by file/commit comparison.
app#122	e85382150853	Case projection/lens surfaces, safe return_to, invitation TTL/CAS/self-member/one-GC constraints, transactional membership, fail-closed Learn, SQL-limited timeline, responsive/a11y tests and Python CI workflow.	P12-02	Retain pending Alan reconciliation; selective port or rebase only after architecture approval	Reinspected at latest head. It is exactly e853821 (no newer delta); retention strengthened, but final extraction waits for Alan checkpoint.

Source	Head	Valuable content	Destination	Recommendation	Extraction status/safety
app#124	84a5a847fea7	Current consolidation of Node 0 questionnaire/guide/surfaces, GC onboarding, participant case shell, invitations, DS assets and large focused test suites.	P12-02	Retain unchanged as evidence; selective port/rewrite after checkpoint	Unextracted; Alan review explicitly incomplete and unresolved.
ds#8	4656426e493e	Consumer parity and bounded app-owned layout guidance for Node 0/case/mobile adoption.	P12-02	Prefer direct review/merge if still correct	Direct-reuse candidate; local dirty asset delta separately protected.

Package source inventory:

- PR #122 responsive journey and lens behavior
- PR #124 Node 0 surfaces
- DS PR #8 parity guidance

1.9 Ownership recommendation

Luke leads this work package, with Alan providing technical review. The split is proposed for review and can be amended before implementation begins.

PR	DRI	Implementer	Reviewer	Final approver	External/high-capability review
P12-01	Luke	Luke (proposed)	Alan	Joint	Mandatory
P12-02	Luke	Luke (proposed)	Alan	Joint	As risk warrants
P12-03	Luke	Luke (proposed)	Alan	Joint	As risk warrants

Luke leads implementation and product acceptance for this package; Alan reviews architecture, migration, security and failure behavior. This is a recommendation pending joint review, not a compulsory assignment.

1.10 Definition of done

- ☐ Every proposed PR is merged in dependency order or explicitly dispositioned, with its objective acceptance criteria and review record complete.
- ☐ All package blocking decisions are resolved by the named owner; no commercial/privacy/security value is guessed in code.
- ☐ Local, PostgreSQL, role/authorization, retry/concurrency, failure, mobile and synthetic-cloud evidence required by this package is attached.
- ☐ Cross-repository contracts and artifact/migration versions are pinned; partial-merge behavior is fail-closed or explicitly degraded.
- ☐ Observability is sufficient to identify revision, transition, dependency and safe failure without sensitive payloads.
- ☐ Existing PR/worktree value and reviewer findings are extracted before closure; no cleanup occurs without team confirmation.
- ☐ The associated programme gate is accepted by Luke and any required external reviewer; tests/deploy alone do not establish B1/B2.